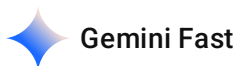


 Example Lecture Scribe (1).pdf L12\_S1\_A.pdf

Using the scribe structure pdf provided in the attached "Example Lecture Scribe" PDF, generate a lecture scribe for the provided Lecture 8 (mention lecture number) materials. the scribe must have all the topics included and it's content based on example pdf. This scribe is intended to serve as a reference material for a closed-notes exam preparation. You must prioritize correctness and maintain strict faithfulness to the provided context, including definitions, notation, theorem statements, and step-by-step proofs (if any ), worked examples exactly as they appear in the slides. Identify every major topic in the lecture, For EACH identified topic, you must independently repeat the given structure in example scribe format. \* Do not include any creative explanations, external intuition, or material not traceable to the attached files\*. Organize the output to follow the logical flow and scope of the lecture to ensure it is suitable for reliable revision.



## CSE 400: Fundamentals of Probability in Computing

[cite\_start]Scribe: Fagun Rathod - AU2440111 [cite: 4] [cite\_start]Lecture: L11\_12: Transformation of Random Variables [cite: 18] [cite\_start]Date: February 10, 2026 [cite: 40, 108]

### [cite\_start]1. Topic Title: Transformation of a Single Random Variable [cite: 44, 60]

#### 2. Definitions and Notation

- [cite\_start]Initial RV:  $X$  with a known PDF  $f_X(x)$  and CDF  $F_X(x)$  [cite: 15, 91].
- [cite\_start]Transformation Function:  $Y = g(X)$  [cite: 35, 61].
- [cite\_start]New RV:  $Y$  [cite: 37].

- [cite\_start]Goal: Find the PDF  $f_Y(y)$  and CDF  $F_Y(y)$ [cite: 39, 51].

### 3. Assumptions / Conditions

- [cite\_start]The function  $g(x)$  is typically assumed to be monotonic (e.g.,  $\text{Log}(\cdot)$ )[cite: 43].
- [cite\_start]The inverse function  $x = g^{-1}(y)$  must exist to solve for  $x$  in terms of  $y$ [cite: 62, 77].

### 4. Main Results / Theorems

The PDF of the transformed variable  $Y$  is given by:  $f_Y(y) = f_X(g^{-1}(y)) \cdot$

$$\left| \frac{d}{dy} g^{-1}(y) \right| \text{ [cite: 66, 68] Alternatively expressed as: } f_Y(y) = \frac{f_X(x)}{\left| \frac{dy}{dx} \right|} \Bigg|_{x=g^{-1}(y)}$$

[cite: 85, 89]

### 5. Proofs / Derivations

- [cite\_start]Step 1: Find the CDF of  $Y$ :  $F_Y(y) = \Pr(Y \leq y) = \Pr(g(X) \leq y) = \Pr(X \leq g^{-1}(y))$ [cite: 74, 75].
- [cite\_start]Step 2: This simplifies to  $F_Y(y) = F_X(g^{-1}(y))$ [cite: 75].
- [cite\_start]Step 3: Differentiate the CDF with respect to  $y$  using the chain rule:  $f_Y(y) = \frac{d}{dy} [F_X(g^{-1}(y))]$ [cite: 64, 66].
- [cite\_start]Step 4: Result:  $f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$ [cite: 68].
- [cite\_start]Note: If  $g(x)$  is decreasing,  $F_Y(y) = 1 - F_X(g^{-1}(y))$ , leading to the absolute value in the general formula[cite: 84, 85, 90].

### 6. Worked Examples

[cite\_start]Example 1: Uniform to Sine Transformation [cite: 94]

- [cite\_start]Given:  $X \sim \text{Uniform}(-1, 1)$ , so  $f_X(x) = \frac{1}{2}$  for  $-1 < x < 1$ [cite: 96].
- [cite\_start]Transformation:  $Y = \sin\left(\frac{\pi X}{2}\right)$ [cite: 97].
- [cite\_start]Step 1 (Inverse):  $x = \frac{2}{\pi} \sin^{-1}(y)$ [cite: 98].
- [cite\_start]Step 2 (Derivative):  $\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$ [cite: 99].
- Step 3 (PDF):  $f_Y(y) = \frac{1}{2} \cdot \left| \frac{2}{\pi \sqrt{1-y^2}} \right|$  [cite\_start] =  $\frac{1}{\pi \sqrt{1-y^2}}$  for  $-1 < y < 1$ [cite: 104, 106, 107].

[cite\_start]2. Topic Title: Function of Two Random Variables ( $Z = X + Y$ ) [cite: 48, 54]

### 2. Definitions and Notation

- [cite\_start]Input RVs:  $X$  and  $Y$  with joint PDF  $f_{XY}(x, y)$  [cite: 137].
- [cite\_start]Transformation:  $Z = g(X, Y) = X + Y$  [cite: 30, 110, 125].
- [cite\_start]Resulting PDF:  $f_Z(z)$  [cite: 112].

#### 4. Main Results / Theorems

If  $X$  and  $Y$  are independent, the PDF of  $Z = X + Y$  is the **convolution** of their individual PDFs: [cite\_start] $f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx$  [cite: 168, 170]

#### 5. Proofs / Derivations

- [cite\_start]Step 1: Write the CDF  $F_Z(z) = Pr(X + Y \leq z)$  [cite: 135].
- [cite\_start]Step 2: Express as a double integral:  $F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{XY}(x, y) dx dy$  [cite: 137].
- [cite\_start]Step 3: Differentiate with respect to  $z$  using the **Leibnitz Rule** [cite: 145, 156]:  
[cite\_start] $\frac{d}{dz} \int_{a(z)}^{b(z)} h(z, y) dy = h(z, b(z)) \frac{db}{dz} - h(z, a(z)) \frac{da}{dz} + \int \frac{\partial h}{\partial z} dy$  [cite: 146, 147].
- [cite\_start]Step 4: Apply Leibnitz to get  $f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(z - y, y) dy$  [cite: 157].

#### 6. Worked Examples

[cite\_start]Example 1: Sum of Independent Normal RVs [cite: 116]

- [cite\_start]Given:  $X \sim N(0, 1)$  and  $Y \sim N(0, 1)$  are independent [cite: 116, 174].
- [cite\_start]Calculation:  $f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \cdot \frac{1}{\sqrt{2\pi}} e^{-(z-x)^2/2} dx$  [cite: 183].
- [cite\_start]Result:  $Z \sim N(0, 2)$  [cite: 117, 191].

[cite\_start]Example 2: Sum of Independent Exponential RVs [cite: 118, 197]

- [cite\_start]Given:  $f_X(x) = \lambda e^{-\lambda x}$  and  $f_Y(y) = \lambda e^{-\lambda y}$  for  $x, y \geq 0$  [cite: 203].
- [cite\_start]Setup:  $f_Z(z) = \int_0^z \lambda e^{-\lambda x} \cdot \lambda e^{-\lambda(z-x)} dx$  [cite: 205].
- **Result:**  $f_Z(z) = \lambda^2 z e^{-\lambda z}$  for  $z \geq 0$  (Erlang distribution).

### [cite\_start]3. Topic Title: Other Joint Transformations ( $Z = X - Y$ and $Z = X/Y$ ) [cite: 31, 34]

#### 2. Definitions and Notation

- [cite\_start]Difference:  $Z = X - Y$  [cite: 225].
- [cite\_start]Ratio:  $Z = X/Y$  [cite: 237].

#### 5. Proofs / Derivations

For  $Z = X - Y$ :

- [cite\_start]Step 1:  $F_Z(z) = Pr(X - Y \leq z) = Pr(X \leq z + Y)$ [cite: 231].
- [cite\_start]Step 2:  $F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z+y} f_{XY}(x, y) dx dy$ [cite: 232].
- [cite\_start]Step 3: Differentiate using the same procedure as the sum[cite: 232].

For  $Z = X/Y$ :

- [cite\_start]Step 1:  $F_Z(z) = Pr(X/Y \leq z)$ [cite: 237].
- [cite\_start]Step 2: Rewrite as  $Pr(X \leq YZ)$  (assuming  $Y > 0$ )[cite: 238].

## 6. Worked Examples

[cite\_start]Example 1: Magnitude of Two RVs [cite: 239]

- [cite\_start]Find: PDF of  $R = \sqrt{X^2 + Y^2}$ [cite: 239].

Would you like me to expand on the Leibnitz Rule steps or provide the specific limits for the joint distribution integrals mentioned in the slides?